CO2 - AT-1

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1) DES vs 3-DES vs Blowfish performance study

To calculate encryption time for DES, 3-DES and Blowfish and compare their security and performance for real-time communication.

Given data :-

Algorithm	key size	Time per block
DES	56 bits	2ms
3-DES	168 bits	6ms
Blowfish	128 bits	1ms

Number of data blocks = 1000

ii) calculate total encryption time

The formula is:

$$T_{total} = T_{Per\ block} \times N$$

where N is the number of blocks

DES (Data Encryption standard)

Given:

* Time per block = 2ms
* Number of blocks = 1000

Calculation :- $T = 2 \times 1000 = 2000ms$

since 1000 ms = 1 second

$T = 2$ seconds

3-DES (Triple Data Encryption standard)

3-DES applies the DES algorithm three times to each block, which generally increase computational cost.

Given,

* Time per block = 6ms
* Number of blocks = 1000

Calculation :- $T = 6 \times 1000 = 6000$ ms

$T = 6$ seconds

Blowfish

Given,

* Time per block = 1ms
* Number of blocks = 1000

Calculation :- $T = 1 \times 1000 = 1000$ ms

$T = 1$ second

Final calculation table

Algorithm	calculation	Total time
DES	2×1000	2 seconds
3-DES	6×1000	6 seconds
Blowfish	1×1000	1 second

~~Recommend the best algorithm for secure real time communication.~~

⇒ Analyze security and performance trade-offs

DES :-

* uses an effective key size of 56 bits
* Has low computational cost compared with 3-DES in the given example
* Its short key makes brute-force attack practical with modern resources

3-DES :-

* Use three DES keys, commonly described as a 168-bit key bundle.
* Applies DES three times, increasing encryption time
* Provides stronger protection than single DES, but its effective security is not equivalent to 168 independent bits.

Blowfish :-

* support variable key lengths; 128 bit is the key size specified in this question
* IS fast in the supplied performance measurements
* Its 64-bits block size limit its suitability for encrypting

large amount of data.

iii) Recommend the best algorithm for secure real-time communication.

=> For real-time communication, an algorithm should provide strong security, low delay, efficient processing and reliable protection of message.

=> Among the three algorithms, Blowfish is the fastest according to the supplied measurement. However, it has a 64-bit block-size limitation.

Recommendation: AES-GCM is a better choice for new real time communication system because:

* AES support 128-bit, 192-bit and 256-bit key
* AES is efficient in both hardware and software
* GCM provides encryption and authentication, helping Protect both confidentiality and integrity.

conclusion :- Blowfish has the lowest calculated time among the given algorithm, but AES-GCM is the recommended practical choice for secure real-time communication.

2) ECC VS RSA Comparison

To compare RSA and ECC based on computational cost, power consumption, memory usage, and suitability for IoT devices.

Given data,

* RSA key size = 1024 bits
* ECC key size = 160 bits
* Device processing capacity = limited
* The question assume equivalent security for comparison.

Note :- RSA-1024 and ECC-160 are too weak for modern security and are not actually equivalent in security strength.

9) Analyze computational cost

RSA (Rivest-Shamir-Adleman)

* Uses Public and private key for encryption, decryption, or digital signature
* Requires large-integer arithmetic and modular exponentiation.
* Private key increase storage and communication requirements.

ECC (Elliptic Curve Cryptography)

* Uses mathematical operations on elliptic curves
* provides strong security with smaller keys than RSA at comparable modern security levels.
* can reduce processing and communication overhead.

by Estimate power and memory consumption

Factor	RSA	ECC
Key size in question	1024 bits	160 bits
Key storage	Larger	smaller
communication overhead	Higher	lower
Memory requirment	Generally higher	Generally lower
IOT suitability	Less suitable	More suitable

Key-size comparison:

$$1024/160 = 6.4$$

=> The RSA key is 6.4 times longer in bit than the ECC key in this example. This does not mean that RSA consume exactly 6.4 times more power or memory.

=> ECC generally reducess communication ollerhead and may reduce energy comsumption, depending on the hardware and implementation.

c) Justify which cryptosystem is better

ECC is generally preferable for resource-limited IOT device

Reasons:-

* smaller key : Require less storage
* Lower communication overhead : less key data needs to be transmitted.
* Efficient processing : can perform public-key operations efficiently on suitable devices.
* Power efficiency : Reduced processing and communication may save energy.
* IOT suitability : works well in systems with limited memory, bandwidth and processing resources.

For real-world applications, modern ECC curves such as P-256 are preferable to ECC-160. RSA-2048 is a more appropriate RSA comparison than RSA-1024 for many current applications.

conclussion :—

ECC is generally more suitable for IoT device because it uses smaller key and often requires fewer computational and communication resources.